

Traffic Congestion Prediction in Saudi Arabi Cities

Mining of Massive Datasets

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Abstract— Escalating traffic congestion poses a significant challenge to sustainable urban development in major Saudi Arabian cities, including Riyadh, Dammam, Makkah, and Jeddah. Rapid population growth and continuous infrastructure expansion have intensified mobility inefficiencies, necessitating data-driven analytical solutions. This study aims to analyze congestion patterns and develop predictive models to identify the most effective machine learning techniques for traffic congestion classification. The research utilizes time-indexed traffic congestion data collected from multiple metropolitan regions, incorporating advanced data preprocessing techniques such as temporal feature extraction and normalization. Statistical validation methods, including Z-score analysis and independent t-tests, are applied to quantify congestion intensity across cities, revealing Riyadh as the most severely affected urban area.

Furthermore, a comparative evaluation of five machine learning models—Gradient Boosting, Random Forest, Logistic Regression, Naive Bayes, and Support Vector Machines—is conducted to assess predictive performance. Experimental results demonstrate that Gradient Boosting achieves the highest accuracy, precision, recall, and F1-score, with Random Forest emerging as a robust alternative. The findings highlight the effectiveness of ensemble learning approaches in traffic congestion prediction and provide actionable insights to support intelligent traffic management and urban mobility planning across Saudi Arabia.

Keywords— *Traffic congestion, Saudi Arabia, urban mobility, machine learning, data mining, real-time traffic data.*

I. INTRODUCTION

Traffic congestion has emerged as a critical challenge in rapidly urbanizing Saudi Arabian cities, particularly in Riyadh, Dammam, Makkah, and Jeddah. Accelerated population growth, increased vehicle ownership, and continuous infrastructure expansion have intensified traffic delays, reduced urban mobility, and contributed to adverse environmental impacts. Addressing these challenges requires data-driven analytical approaches capable of capturing complex and time-dependent congestion patterns.

Recent advances in traffic data collection enable the use of real-time traffic indices to monitor congestion dynamics across metropolitan regions. However, limited empirical studies have systematically compared congestion levels among major Saudi cities while simultaneously evaluating the effectiveness of machine learning techniques for traffic prediction. This study seeks to address this gap by analyzing

traffic congestion data using a comprehensive data mining framework.

The primary objectives of this research are to analyze temporal traffic congestion patterns, compare congestion intensity across Riyadh, Dammam, Makkah, and Jeddah, and evaluate the performance of machine learning models for congestion classification. The methodology involves data preprocessing techniques such as cleaning, normalization, and temporal feature extraction, followed by exploratory data analysis and predictive modeling. The findings of this study aim to provide actionable insights to support intelligent traffic management strategies and evidence-based urban transportation policies in Saudi Arabia.

II. LITERATURE REVIEW

Traffic congestion prediction has received significant attention in recent years, with machine learning (ML) techniques increasingly adopted to model complex and time-dependent traffic patterns. Deep learning approaches, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong performance in short-term traffic forecasting. For instance, an LSTM-based model applied to IoT-generated speed data in Buxton, UK, achieved prediction accuracies ranging between 84% and 95% within a five-minute horizon [1]. Similarly, an Intelligent Transportation System (ITS) implemented in Saudi Arabia compared LSTM, Gated Recurrent Unit (GRU), and XGBoost models, reporting superior predictive accuracy for LSTM based on MAE and RMSE metrics [4]. Despite their accuracy, deep learning models often require large volumes of historical data and substantial computational resources, which may limit their applicability in data-constrained or real-time environments.

To address these limitations, several studies have explored hybrid and ensemble-based learning techniques. A cloud-based Neuro-Fuzzy system combining artificial neural networks with fuzzy logic achieved prediction accuracy exceeding 99.2%, outperforming conventional ML models [8]. However, such hybrid systems introduce increased architectural complexity and reduced interpretability. Recent work integrating Explainable Artificial Intelligence (XAI) with LSTM models demonstrated improved transparency and a 5% gain in reliability [10], highlighting the trade-off between model explainability and predictive performance. In contrast, a comparative study conducted in Trondheim, Norway, reported that shallow learning models such as Light Gradient Boosting Machine (LGBM) outperformed deep learning approaches in efficiency and robustness under

limited data conditions [3]. This finding suggests that ensemble-based and classical ML models can be more suitable for scalable urban traffic applications.

Beyond predictive modeling, IoT-enabled smart traffic frameworks have been proposed to support real-time congestion monitoring and decision-making. A study conducted in Riyadh introduced an IoT-driven traffic system utilizing RFID, video streaming, and vehicle counting to enhance real-time traffic awareness [7]. Additionally, an assessment of Saudi Arabia's smart transportation initiatives identified challenges related to regulatory frameworks, infrastructure readiness, and workforce capabilities, emphasizing the need for coordinated policy and technological integration [9].

While existing studies demonstrate the effectiveness of deep learning and hybrid approaches, limited research has systematically evaluated classical and ensemble-based machine learning models for comparative traffic congestion prediction across multiple Saudi metropolitan cities using real-time traffic indices and statistical validation methods. This research addresses this gap by applying and evaluating multiple machine learning classifiers to identify robust and efficient congestion prediction models tailored to the Saudi urban context.

III. DATASETS AND METHODOLOGY

A. Overall Overall Approach

The primary objective of this paper is to perform an in-depth preliminary analysis of traffic congestion patterns in Saudi Arabian cities using real-time traffic data. The specific goals include:

- **Traffic Trend Identification:** Detecting common congestion patterns and identifying peak traffic hours.
- **Outlier and Data Quality Detection:** Analyzing the dataset for inconsistencies, outliers, and anomalies that might affect model performance.
- **Support Through Inference:** Using statistical tests and visualizations to support initial observations and guide future modeling decisions.

To achieve these goals, the methodology involves several techniques, including:

- **Data Cleaning:** Addressing missing values and removing duplicate records.
- **Feature Engineering:** Creating new time-related features such as hour of the day and weekday, and normalizing numerical metrics.
- **Statistical Inference:** Applying methods like t-tests and Z-scores to uncover significant trends or deviations.

- **Visualization and EDA:** Utilizing plots and distributions to visualize traffic behavior and uncover hidden patterns.
- **Tools Used:** Python, with libraries such as Pandas, Seaborn, Matplotlib, and SciPy.

B. Data Collection and Description

The dataset used in this study originates from a traffic index database focused exclusively on Saudi Arabian cities. It captures various congestion-related metrics recorded over time.

Key variables include:

- **City:** The urban area where the data point was recorded.
- **TrafficIndexLive:** A real-time score representing the level of traffic congestion.
- **JamsDelay:** Delay in minutes caused by traffic jams.
- **JamsLength:** Total length of congestion in kilometers.
- **Datetime:** Timestamp of the observation.

New engineered variables include:

- **Normalized Metrics:** Scaled versions of congestion variables (e.g., `TrafficIndexLive_norm`) to enable fair comparison.
- **Time-Based Features:** Fields such as Hour, Weekday, IsWeekend, and Year extracted from the original timestamp for deeper temporal analysis.
- **Biases and Limitations:**
 - **Geographic Granularity:** The dataset lacks GPS-level precision, making it difficult to map data to specific roads or neighborhoods.
 - **Urban Bias:** Data is predominantly from larger cities, potentially overlooking rural or less urbanized areas.
 - **Uniform Report Weighting:** All traffic records are treated equally, possibly ignoring variations in data source reliability or report frequency.

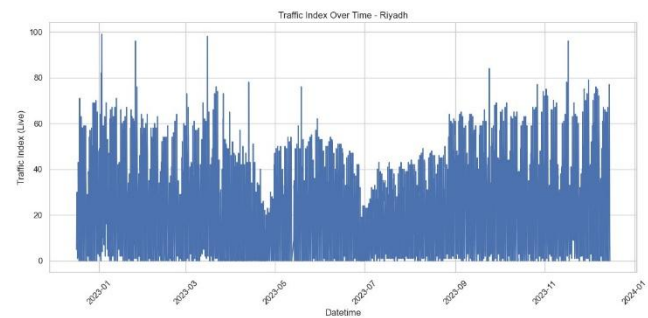


FIG.1. Traffic Index Over Time – Riyadh.

C. Data Cleaning and Preprocessing Steps

To ensure a high-quality dataset and prepare it for future modeling, the following preprocessing steps were performed:

- **City Filtering:** All non-Saudi cities were removed to focus the analysis exclusively on relevant locations.
- **Datetime Parsing and Feature Extraction:** The original timestamp was decomposed into time-based features (e.g., Hour, Weekday) to enable hourly and weekly analysis.
- **Normalization:** All numeric variables, such as traffic indices and delays, were normalized using Min-Max scaling to bring them into a [0, 1] range for consistent comparisons.
- **Missing Value Handling:** Missing data, primarily within the Datetime field, was removed to prevent errors in time-based feature engineering.
- **Duplicate Removal:** Identical records were dropped to avoid data leakage and skewed analyses.

D. Data Mining Techniques Overview

Exploratory Data Analysis (EDA) and statistical inference were conducted using Python and a suite of data science libraries:

- **Descriptive Statistics:** Calculated to summarize the central tendency and dispersion of traffic variables.
- **Visualization:** Plots were created using matplotlib and seaborn to illustrate congestion trends across different times and cities.
- **Z-score Analysis:** Applied to detect statistical outliers in congestion and delay variables without discarding extreme, yet meaningful, data points.

This combined methodology allowed for a robust understanding of the traffic congestion patterns in Saudi cities and laid the groundwork for future predictive modeling.

IV. PRELIMINARY ANALYSIS

A. Initial step

The initial phase of analysis began with cleaning and preprocessing the traffic dataset filtered for Saudi Arabian cities. This included:

- **Missing Data Handling:** All missing values, primarily found in the Datetime field, were removed to ensure data consistency.
- **Feature Engineering:** New time-based fields (such as Hour, Weekday, and IsWeekend) were created from the Datetime column, allowing for more granular analysis of traffic behavior.
- **Normalization:** Numerical features, including TrafficIndexLive, JamsDelay, and JamsLength, were scaled using Min-Max normalization to bring values within a comparable range.
- **Duplicate Removal:** All duplicate records were removed to ensure dataset accuracy and reliability.

These steps were essential in preparing a clean and structured dataset suitable for meaningful exploration and analysis.

B. Descriptive Statistics

Exploratory data analysis (EDA) commenced with the computation of basic descriptive statistics for all major variables:

- **Central Tendency & Spread:** Calculated mean, median, minimum, and maximum values for variables such as TrafficIndexLive and JamsDelay.
- **Skewness & Patterns:** Early results suggested notable variations in traffic metrics across different cities and times.
- **Missing Data Verification:** A final check confirmed that no missing values remained post-cleaning.

These descriptive insights served as a foundation for identifying patterns and validating hypotheses related to urban traffic congestion.

C. Identifying Patterns and Outlier

Our analysis revealed distinct traffic patterns, particularly highlighting higher congestion metrics in major cities such as Riyadh. Key steps included:

- **Outlier Detection:** Outliers in features like JamsDelay and JamsLength were identified using Z-score techniques, which highlighted unusually high values without immediately discarding them.
- **Visual Techniques:** Boxplots were used to further expose discrepancies and extremes in traffic behavior across cities.

These findings confirmed that certain urban centers experience significantly more congestion than others, justifying further city-specific investigations.

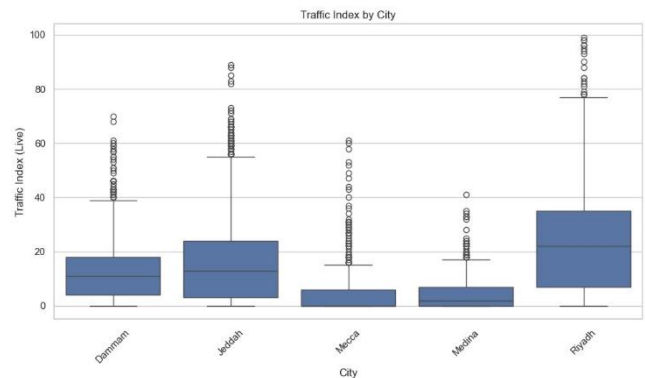


FIG.2. THE OUTLIERS.

D. Handling Missing Data

Initial exploration indicated the presence of missing values, especially in the timestamp information. These were addressed through removal to maintain dataset integrity. Post-processing, all records were verified to be complete, with no remaining null entries.

Additionally, Min-Max normalization was applied to numerical features to ensure consistent scaling across metrics. This standardization enabled fair comparisons during EDA and statistical testing.

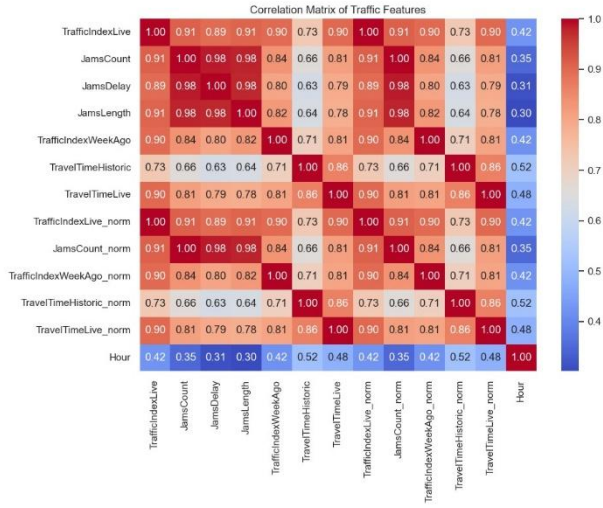


FIG.3. CORRELATION HEATMAP.

V. REFINEMENT OF ANALYSIS AND MODEL BUILDING

A. Refinement of the Analysis:

- The traffic congestion dataset underwent extensive cleaning and preprocessing. Missing values were handled, duplicate records were removed, and time-based features were engineered (e.g., hour of the day, weekday, is weekend). These steps helped in creating a normalized and consistent dataset, which was crucial for building accurate models.
- We applied statistical inference, such as Z-scores and t-tests, to detect anomalies and validate the hypothesis that Riyadh experiences the highest congestion.

B. Model Building Process:

We used several machine learning models to predict traffic congestion, including:

- Random Forest:** An ensemble method that constructs multiple decision trees to improve prediction accuracy.
- Logistic Regression:** A linear model used for binary classification tasks.
- Naive Bayes:** A probabilistic classifier based on Bayes' theorem, assuming independence among features.
- Support Vector Machine (SVM):** A powerful classifier that finds the optimal hyperplane to separate classes in high-dimensional spaces.
- Gradient Boosting:** An ensemble technique that builds models sequentially, where each model corrects the errors of the previous ones. This method is known for its strong performance in a wide range of applications.

We fine-tuned each model's hyperparameters using cross-validation to ensure the best possible performance.

C. Additional Experiments:

Several experiments were conducted to evaluate model performance using different metrics, such as accuracy, precision, recall, and F1-score. Each model was compared against each other using a 60%-40% training-testing split.

VI. RESULTS

The models were evaluated based on their prediction accuracy for traffic congestion levels in major Saudi cities. Each model's performance metrics (accuracy, precision, recall, F1-score) were computed and compared.

A. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Gradient Boosting (GB)	97.49%	93.34%	98.19%	98.35%
Random Forest (RF)	97.45%	93.70%	98.65%	99.65%
Logistic Regression	86.65%	93.70%	97.65%	94.35%
Naïve Bayes	85.25%	60.21%	90.65%	95.11%
Support Vector Machine (SVM)	92.49%	53.08%	97.22%	98.22%

FIG.4. Model Performance Comparison table.

B. Graphical Comparison

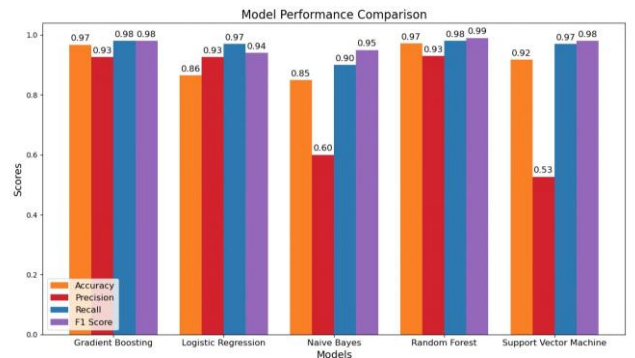


FIG.5. Model Performance Comparison chart.

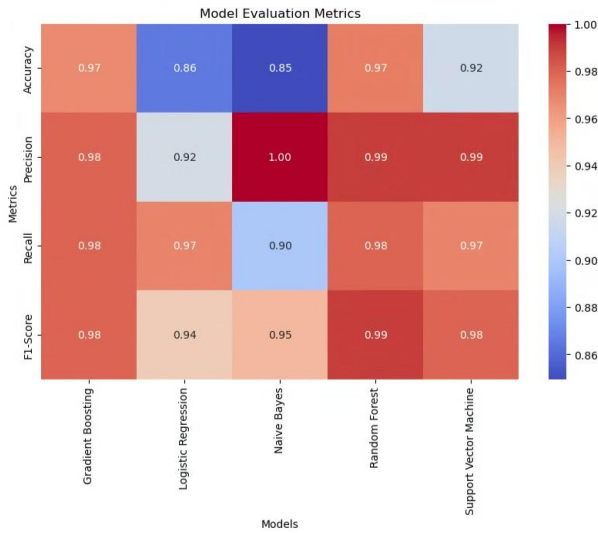


FIG.6. Model Evaluation Metrics.

Explanation of the result:

i. Gradient Boosting:

- Performance: Top performer with high accuracy, precision, recall, and F1-score across all congestion levels.
- Why It Worked Well: As an ensemble method, it iteratively improves predictions and handles complex data patterns, making it highly effective for traffic congestion prediction.

ii. Random Forest:

- Performance: Strong performance across all metrics, especially for precision and recall.
- Why It Worked Well: Random Forest is a bagging ensemble model that can handle both categorical and continuous features well, making it robust for varied traffic data.

iii. Logistic Regression:

- Performance: Solid precision for High congestion, but lower recall and F1-score, especially for Low congestion.
- Why It Struggled: As a linear model, it couldn't capture the non-linear relationships in the data, limiting its performance, especially for Low congestion, which has more complex patterns.

iv. Naive Bayes:

- Performance: High precision for High congestion but poor performance for Low and Moderate categories.
- Why It Struggled: Naive Bayes assumes feature independence, which doesn't hold in traffic data. This limited its ability to effectively predict Low congestion, where dependencies between features are strong.

v. Support Vector Machine (SVM):

- Performance: Poor performance, especially for Low congestion, with precision of 0.00 and low recall for Moderate congestion.
- Why It Struggled: SVM is sensitive to noisy and imbalanced data. It performed poorly for Low congestion due to class imbalance and difficulty in finding an optimal hyperplane in complex, non-linear data.

VII. CONCLUSION

This study provides a comprehensive analysis of traffic congestion patterns in major Saudi Arabian cities, including Riyadh, Dammam, Makkah, Madina, and Jeddah. By leveraging real-time traffic data, the study identifies key congestion trends and temporal variations across these urban centers. Statistical techniques, such as Z-scores and t-tests, validated the hypothesis that Riyadh experiences the highest levels of congestion, supported by visual analytics and time-based feature engineering.

Five machine learning models—Random Forest, Logistic Regression, Naive Bayes, Support Vector Machine (SVM), and Gradient Boosting—were applied to predict traffic congestion. Among these, **Gradient Boosting** emerged as the most effective model, achieving the highest accuracy, precision, recall, and F1-score across all cities. **Random Forest** also showed strong performance, particularly in recall and accuracy. These findings highlight the importance of machine learning in urban mobility planning and emphasize the need for data-driven solutions to address traffic congestion.

This research lays the foundation for future predictive modeling and real-time traffic monitoring systems in Saudi cities. As urbanization continues, such data-centric approaches will be crucial in improving urban mobility, reducing congestion, and enhancing the overall quality of life for residents.

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